Measuring Fairness in Energy Management Algorithms

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***Abstract*—This paper examines fairness in demand side management by applying different fairness indices to the results of the profile steering algorithm. The outcomes provided by these indices are evaluated and assessed on how effectively they indicate fairness. After analysing Jain's index, the Gini coefficient, and the quality of experience fairness index, both Jain's index and the Gini coefficient show similar results and can be used to improve profile steering. Possibilities were explored to improve fairness in profile steering using these indices.**

***Index Terms*—profile steering, peak shaving, fairness indices, energy storage system flexibility**

I. Introduction

The modern energy grid faces an increasing amount of strain due to the growing use of devices such as heat pumps, solar panels, and electric vehicles. This causes unwanted peaks in both electricity production and consumption, which leads to grid congestions. A way to reduce these peaks is to use Demand Side Management (DSM). DSM changes the behaviour of users to better align with a desired load profile.

One way in which DSM is applied is by using the profile steering (PS) algorithm proposed in [1]. This algorithm analyses and subsequently utilises the offered flexibility from each household by time-shifting and/or buffering the energy consumption of devices within households. The PS algorithm does, however, prioritise the usage of the household that provides the most flexibility. This can be seen as unfair due to unequal utilization of resources. The main goal of this paper is to find a way to measure fairness in PS in order to make it more fair.

Among these various flexibility sources, energy storage systems (ESSs) offer the greatest flexibility. Therefore, this paper focuses solely on assessing the fairness of their utilization.

In this paper, section II gives an overview of PS and fairness indices. These fairness indices require an input, which will be called the fairness input. Section III explains how the data is converted into the fairness inputs used in the analysis, followed by the development of an ESS fairness simulation in section IV. The observations from the simulations are then presented in section V, leading to the conclusion in section VII. Finally, section VIII provides recommendations for improving fairness in PS.

II. Background

This section introduces PS and explores fairness indices as tools to measure the fairness of ESS usage.

A. Profile steering

The goal of the PS algorithm is to match energy usage with a target profile to reduce power peaks and improve grid stability. The algorithm uses an iterative process to reach the target profile. In each step, it evaluates how each device can contribute to aligning the current power profile with the target profile. The device that provides the best improvement is selected, and its adjustments are integrated into the overall profile. This iterative process always happens in the same order. As a consequence, if there are multiple households offering the same amount of flexibility, the first one in the PS algorithm's list is always used first [1].

B. Fairness indices

Fairness indices are mathematical tools used to determine how evenly resources are distributed, handled, or perceived in a system. This section explains three fairness indices: Jain's Index, the Gini coefficient, and the QoE fairness index.

1) Jain's index: Jain's Index is a widely used metric to measure fairness in resource allocation. It is commonly applied in computer networking and telecommunications to ensure fair distribution of bandwidth among multiple users. The index ranges from 0 to 1, where 1 indicates perfect fairness (all entities receive equal resources), and values closer to 0 indicate significant inequality [2].

$$J = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \cdot \sum_{i=1}^{n} x_i^2} \tag{1}$$

Equation (1) shows the formula for Jain's index [2]. Here, n represents the number of users, and x_i is the resource allocation for user i.

It effectively identifies imbalances in resource allocation and works well for systems with many users or entities [2]. Being widely used, Jain's index has been the subject of much research. However, it assumes equal distribution is ideal, so it does not account for situations where users may have different needs or priorities [3].

2) Gini coefficient: The Gini coefficient is a metric to measure inequality, which is commonly used in economics. Its output ranges between 0 and 1, where a value of 0 means that everything is perfectly fair, and a value of 1 means that one person has everything and everyone else has nothing [4].

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n^2 \bar{x}} \tag{2}$$

The formula for the Gini coefficient can be found in Equation (2) [4]. Here, n represents the number of elements in the dataset, x is the set of all input values, and $\bar{x}$ the average input value. Since the result of the Gini coefficient is a metric for unfairness, the formula $1 - G$ is used to measure fairness instead.

The Gini coefficient is an effective metric for measuring overall inequality [5]. However, it lacks sensitivity when dealing with small datasets [6].

3) QoE fairness index:

The quality of experience (QoE) fairness index proposed by Hoßfeld et al. provides a user-centric measure of fairness in QoE management [7].

$$F = 1 - \frac{2\sigma}{H - L} \tag{3}$$

Equation (3) is the formula for the QoE fairness index. Here σ is the standard deviation of the QoE values for users, and H and L are the upper and lower bounds of the QoE scale, respectively.

The QoE fairness index provides a measure of fairness in a QoE distribution, with $F \in [0, 1]$, where $F = 1$ indicating perfect fairness [7]. Unlike other metrics, such as Jain's fairness index, the QoE index is specifically designed to handle interval scales rather than ratio scales. It also requires an upper and lower bound, which the other indices don't need.

III. Fairness input

To be able to calculate fairness in energy management, the discussed fairness functions require an input. This fairness input f is a number calculated for each household, and should represent how much a household's ESS has been used. This can be done by adding the sum of all the energy that has entered the ESS to the sum of all the energy that has left the ESS, during a time period T. The result is the energy usage of the ESS, E_u. Given the power profile P_i of the i-th ESS, its usage $E_{u,i}$ can be calculated by using eq. (4).

$$E_{u,i} = \sum_{t \in T} |\tau P_{i,t}| \tag{4}$$

In eq. (4), t is a time interval within T, τ is the duration of such an interval, and $P_{i,t}$ is the charging or discharging power of the i-th ESS during interval t.

E_u is suitable to use as a fairness input. It could, however, lead to situations where different households have drastically different usage values, purely because their ESS capacities differ. To combat this, $E_{u,i}$ can be divided by the capacity of its respective ESS: $ESS_{cap,i}$. This way, the fairness input becomes proportional to the ESS capacity of the house.

This results in two possible fairness inputs: a normal fairness input f and a proportional fairness input f_p. These are both defined in eqs. (5) and (6), respectively.

$$f_i = E_{u,i} \tag{5}$$

$$f_{p,i} = \frac{E_{u,i}}{ESS_{cap,i}} \tag{6}$$

IV. ESS fairness simulation

To test the fairness of PS, a number of simulation were created. These simulations use data from an energy community called Aardehuizen, located in Olst, The Netherlands. This community consists of 23 eco-friendly homes [8]. The electricity usage of each of these homes was collected at a 15 minute interval over the course of a year. For the simulation, each house is equipped with a battery, a heat pump, and an electric vehicle. All these devices have identical specifications. This ensures that differences in simulation results are not influenced by variations in these devices.

A number of simulation setups were created in order to properly determine the behaviour and fairness of PS. These simulations are:

TABLE I
Simulations

Nr	Name	Description	Settings
1	All-equal Under-capacity	Too little battery capacity to balance the grid effectively	capacity=400Wh
2	All-equal Over-capacity	More than enough battery capacity to balance the grid	capacity=25kWh
3	All-different	All batteries have a different capacity, following a step function	start size=1kWh, step size=500Wh
4	One-different	All batteries have the same capacity except for one battery	capacity=1kWh, outlier capacity=6kWh

In the simulations, the PS algorithm optimizes the power usage of each device to match the ideal power profile as closely as possible. f and f_p are then calculated from these profiles using the fairness input formula eq. (5). These inputs are then used to calculate the fairness, using eqs. (1) to (3).

V. Observations

This section discusses the results of the simulations. It describes why QoE is not suitable for PS and analyses the results of the simulations.

A. QoE index analysis

The QoE index requires upper and lower bounds. ESSs have a maximum charging rate, so the fairness inputs do have a maximum. This maximum could be used as the upper bound, with 0 being the lower bound. However, this maximum will often be much larger than the actual usage values. Using this maximum causes the QoE index to deem unfair situations fair, because the input values are not far apart compared to its bounds.

Another option could be to use bounds based on the maximum and minimum values of the inputs. This does not provide a useful output either, however, since this results in two of the values always being maximally far apart.

Figure 1 shows the result of the fairness indices when changing one of the inputs (the x-axis). The other three inputs are indicated with red ticks on the x-axis. The QoE index clearly produces unusable results in both cases.

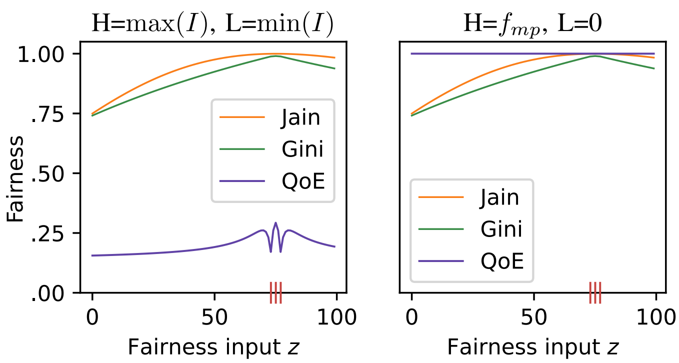

Fig. 1. Two graphs showing two possible ways of defining bounds for the QoE index. Here I is the set of all fairness inputs and f_{mp} is the maximum *possible* fairness input, described in section V-A.

Since the QoE is not fit to measure fairness in PS, only Jain's index and the Gini coefficient will be used from this point.

B. Simulation results

All simulations described in table I were performed. The results are described in the following sections.

1) Over-capacity and under-capacity: In the first two simulations listed in table I, all batteries have the same capacity. The only difference between the two simulations is that in simulation 2 the batteries have more than enough capacity to balance the grid, whereas in simulation 1 the total capacity is insufficient to balance the grid. Figures 2 and 3 show the results of the two simulations. Here it can be seen that in the under-capacity scenario PS behaves more fair. This is because in this scenario, PS requires every battery to be used. In the over-capacity simulation PS behaves less fair. In this simulation PS only needs a few batteries to balance the grid, leading to the first batteries being used excessively compared to the others.

2) All-different: The results of simulation 3 from table I are shown in figs. 4 and 5. One noticeable trend is that with the exception of the biggest few batteries, the usages are all proportional to their capacity. Because the usage of most batteries is proportional to their capacity, this simulation has a much higher proportional fairness value compared to normal fairness.

3) One-different: One interesting observation is that Jain's and Gini's indices generally align. An exception to this appears to occur when some batteries have a significantly higher usage than the others. This was simulated in simulation 4 of table I. The results of this simulation are shown in fig. 6.

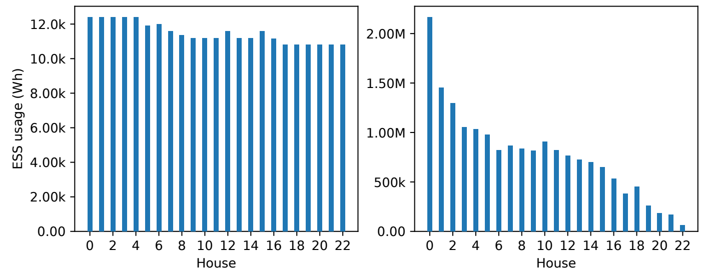

Fig. 2. The sum of the battery usage of every house, measured during the under-capacity (left) and over-capacity (right) simulations, over the span of 31 days.

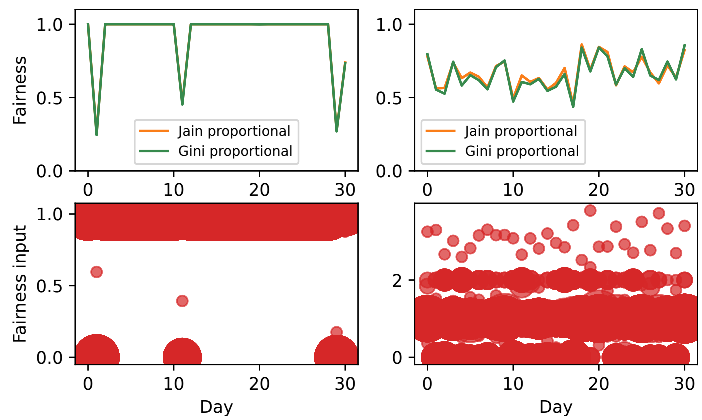

Fig. 3. The measured fairness of the under-capacity (left) and over-capacity (right) simulations, with the respective fairness inputs pictured below that. A larger red circle means more inputs have the same value or are very close to that value.

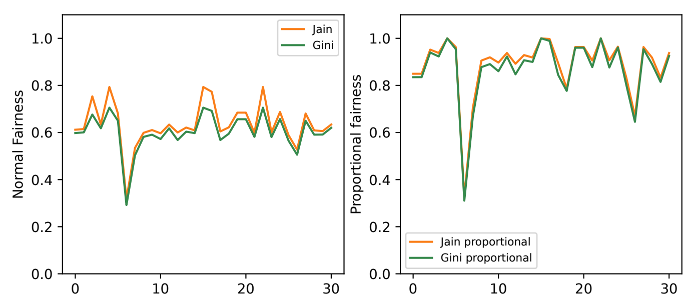

Fig. 4. The fairness results of the all-different simulation.

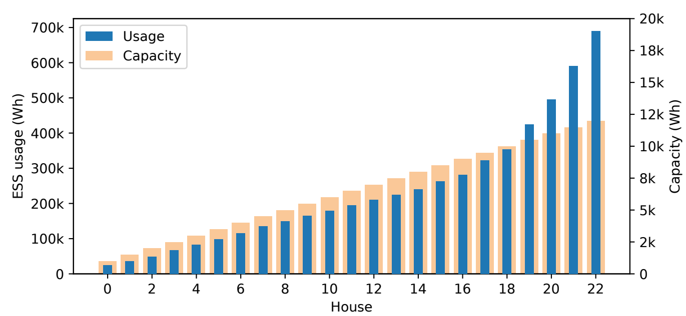

Fig. 5. The sum of the battery usage together with the battery capacity of each house, measured during the all-different simulation.

The significant difference in capacity creates a noticeable difference between the two fairness indices.

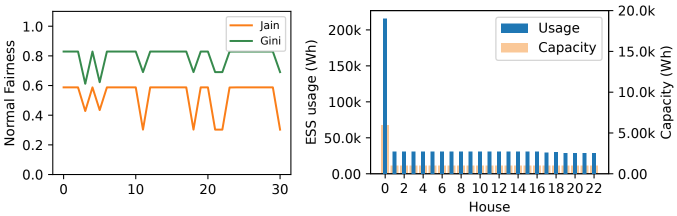

Fig. 6. Figure of the normal fairness results (left) and battery usage (right) for the one-different simulation over the span of 31 days.

VI. Improving Fairness in Profile Steering

The results of the simulations show that PS is not a fair algorithm. There are numerous possible ways to potentially improve the fairness of the algorithm. This section describes two methods of doing so.

A. *Shuffling batteries in profile steering*

Due to the iterative nature of PS, when using batteries that have the same size, the first battery in the list is always chosen first. One way to improve fairness is by using a shuffling mechanism. This shuffle ensures that, in each iteration, the order of devices is randomized before they propose updates to the profile. When multiple batteries are of equal capacity, this results in a different battery being used excessively every day. This means that on a day-to-day basis the fairness is still low, but when averaged over multiple days, the fairness improves significantly. This can be seen in figs. 7 and 8, where both shuffled and unshuffled results are shown using a rolling average of 7 days to calculate the fairness. Shuffling only has an effect when multiple batteries have the same capacity, so its application is limited.

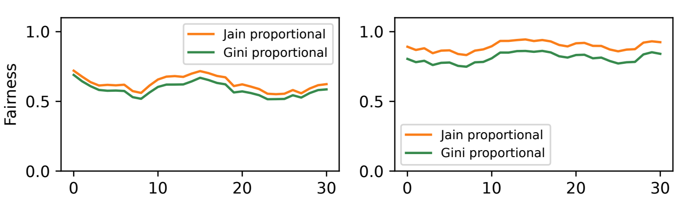

Fig. 7. The fairness results of the shuffled (right) and unshuffled (left) over-capacity simulation with a rolling average of 7 days, over the span of 31 days.

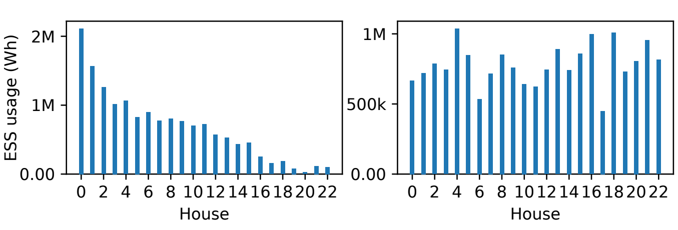

Fig. 8. The battery usages of the shuffled (right) and unshuffled (left) over-capacity simulation over the span of 31 days.

B. *Using fairness indices in profile steering*

Another way to improve the fairness of PS is to change the way it chooses which device to use in improving the profile. The PS algorithm uses the euclidean distance between the new profile and the target profile as metric to choose a device. To make the algorithm more fair, this metric could be extended to also consider how choosing a particular device impacts the overall fairness of the profile.

One way of combining these two metrics is to use a linear combination. This only applies to ESSs; the improvement of other devices is calculated normally. This method, and a number other methods were applied to the algorithm and subsequently tested. The tests are listed here:

- A linear combination of the fairness improvement and the profile improvement, using numerous different combinations of constants.
- Using the product of the profile improvement and the new fairness value the profile would result in as the improvement value for batteries.
- Subtracting the usage multiplied by a constant factor from the profile improvement to discourage using batteries that are already being used.

These simulations all either result in no significant improvement in fairness whatsoever, or result in no ESSs being used at all. Due to time constraints, this subject was not explored in more detail. More research still needs to be done on this subject.

VII. Conclusion

Profile steering does not treat the use of energy storage systems fair because it always uses the biggest source of flexibility first. Jain's and Gini's fairness indices are effective tools for measuring fairness in profile steering. Using normal fairness inputs is more useful when the main priority is the total usage, whereas proportional fairness inputs are desired when the capacities of the batteries also need to be taken in to account. Further integrating fairness metrics into the algorithm offers a path toward more balanced energy management. Shuffling the order of devices in profile steering improves the fairness, but only if most devices have the same capacity.

VIII. Recommendations

This paper recommends using either Jain's index or the Gini coefficient to assess the fairness of the PS algorithm, excluding QoE as it is not applicable to PS. Due to time constraints, there was not enough time to explore the application of fairness indices in PS in detail. Future works could expand on this by incorporating fairness indices directly into the PS algorithm in order to make its results more fair.

References

[1] M. E. T. Gerards, H. A. Toersche, G. Hoogsteen, T. van der Klauw, J. L. Hurink, and G. J. M. Smit, "Demand side management using profile steering," in *2015 IEEE Eindhoven PowerTech*, 2015, pp. 1–6.

[2] R. K. Jain, D.-M. W. Chiu, W. R. Hawe *et al.*, "A quantitative measure of fairness and discrimination," *Eastern Research Laboratory, Digital Equipment Corporation, Hudson, MA*, vol. 21, p. 1, 1984.

[3] A. F. Rochim, A. Muis, and R. F. Sari, "A discrimination index based on jain's fairness index to differentiate researchers with identical h-index values," *Journal of Data and Information Science*, vol. 5, no. 4, pp. 5–18, 2020.

[4] A. Druckman and T. Jackson, "Measuring resource inequalities: The concepts and methodology for an area-based Gini coefficient," *Ecological Economics*, vol. 65, no. 2, p. 242–252, Apr 2008.

[5] V. Xinying Chen and J. N. Hooker, "A guide to formulating fairness in an optimization model," *Annals of Operations Research*, vol. 326, no. 1, pp. 581–619, 7 2023.

[6] A. Jacobson, A. D. Milman, and D. M. Kammen, "Letting the (energy) Gini out of the bottle: Lorenz curves of cumulative electricity consumption and Gini coefficients as metrics of energy distribution and equity," *Energy Policy*, vol. 33, no. 14, pp. 1825–1832, 2005.

[7] T. Hoßfeld, L. Skorin-Kapov, P. E. Heegaard, and M. Varela, "A new QoE fairness index for QoE management," *Quality and User Experience*, vol. 3, no. 1, p. 4, 2 2018.

[8] Aardehuizen Olst, "Aardehuizen olst - duurzaam wonen in aardehuizen," https://www.aardehuis.nl/, accessed: 2025-01-09.